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Cassette and apparatus for packing disposable objects into an elongated tube of flexible material

Abstract

A cassette for dispensing bags from an elongated tubing comprising an annular receptacle accommodating a length of tubing in an accumulated condition. An annular opening at an upper end of the annular receptacle is for dispensing the tubing. The annular receptacle defines a central opening through which a knotted end of the tubing passes to form a bag supported by the annular receptacle with disposable objects passing through the circular central opening to be received in the bag, and a clearance only at a bottom of the central opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present patent application is a continuation of U.S. patent application Ser. No. 15/145,977 filed on May 4, 2016, which is a continuation of U.S. patent application Ser. No. 14/526,650 filed on Oct. 29, 2014, now abandoned, which is a continuation of U.S. patent application Ser. No. 14/017,842 filed on Sep. 4, 2013, now patented to U.S. Pat. No. 8,899,420, which is a continuation of U.S. patent application Ser. No. 13/324,234 filed on Dec. 13, 2011, now abandoned, which is a divisional of U.S. patent application Ser. No. 12/245,172, filed on Oct. 3, 2008, now abandoned, which claims priority to European Patent Application No. 07019571.4, filed on Oct. 5, 2007, all of said applications being incorporated by reference herein in their entirety.

FIELD OF THE APPLICATION

(1) The present application relates to an apparatus for packaging disposable material or objects into a tube of flexible plastic film material. More specifically, the invention relates to an apparatus for use to package waste material, for example babies' disposable diapers, or any other kind of appropriate objects or material, into a tube of flexible plastic film material and to store the so packaged waste material in a hygienic and a substantially odor-free manner, until it is collected.

BACKGROUND OF THE ART

(2) Known apparatuses of the above-mentioned type generally comprise a container having an open or openable upper portion in which the waste to be disposed of may be inserted and a bottom portion in which the disposed waste is stored. A ring-shaped cassette is mounted in the upper portion of the container, and contains, in a pleated form, a tube of flexible plastic film material which is formed into an elongated tube which is knotted at a bottom end and into which the waste material may be inserted and temporarily stored.

(3) In use, the waste to be disposed of is inserted into the tube at the upper portion of the container and the

tube and waste are then pushed through the open center of the cassette towards the bottom portion of the container for storage purposes. Closeable means are also provided for closing the tube below the cassette and thus preventing bad odors from escaping from the tube during storage.

(4) In Canadian Patent No. 1,298,191, these closeable means are disclosed as a core that can be turned by a lid about a cylinder in order to twist the tube at regular intervals to form successive "pouches" that are kept sealed while they are stored.

(5) In Canadian laid-open Application No. 2,383,799, a squeezing device is mounted in the container below the cassette to pull the tube from the cassette and move it down together with the objects inserted into the container. This squeezing device comprises a pair of opposite rotatable members between which the tube is inserted. The rotatable members have a plurality of opposite bars extending transversely to the tube in order to squeeze the tube, keep it closed until other objects to be disposed of are inserted into the upper portion of the container, and pull it down to move the plastic film forming the tube and the objects contained therein towards the bottom portion of the container for storage purpose. Actuation of the rotatable members in unison and in opposite direction to achieve the requested squeezing, closing and pulling down of the plastic film tube is preferably obtained by actuation of a lever that is part of the container.

(6) In Canadian laid-open Application No. 2,441,837, a plunging device is provided to compress the object to be disposed of, and push it into the tube and to the bottom part of the container. The plunging device includes two arms with pivoting flaps attached thereto and opposing pivotable slides.

(7) All of these prior art devices have several disadvantages. Their mechanisms have many parts and are prone to breakage. These devices are also not user-friendly and difficult to understand the operation thereof when first purchased. They are costly to fabricate and they utilize excess film from the cassettes, adding to the cost of use. Some also do not effectively contain the odors of the disposed material. Some cause confusion in their installation, causing improperly oriented cassettes and contamination of the cassette walls.

SUMMARY OF THE APPLICATION

(8) It is therefore an aim of the present invention to provide a novel apparatus for packing disposable objects into an elongated tube of flexible plastic film material that addresses issues associated with the prior art.

(9) Therefore, in accordance with a first embodiment of the present application, there is provided a cassette for dispensing bags from an elongated tubing comprising an annular receptacle accommodating a length of tubing in an accumulated condition, an annular opening at an upper end of the annular receptacle for dispensing the tubing, the annular receptacle defining a central opening through which a knotted end of the tubing passes to form a bag supported by the annular receptacle with disposable objects passing through the circular central opening to be received in the bag, and a clearance only at a bottom of the central opening.

(10) In accordance with the first embodiment, the clearance is in the shape of a chamfer.

(11) In accordance with the first embodiment, a removable lid closes the annular opening of the annular receptacle.

(12) In accordance with the first embodiment, an outer periphery of the annular receptacle tapers downwardly.

(13) In accordance with the first embodiment, the annular opening is adjacent to an outer periphery of the annular receptacle.

(14) In accordance with the first embodiment, the clearance extends on the full periphery of the central opening.

(15) In accordance with the first embodiment of the present application, there is provided an apparatus for packing at least one disposable object into an elongated tube of flexible material, comprising a bin defining an enclosure with an opening defined at a top thereof for receiving at least one disposable object to be packed; a holder within the bin proximate the opening; a closing mechanism located below the holder comprising first and second portions, the second portion being biased in a closed position against the first portion and defining a receiving surface for receiving the at least one disposable object, the second portion being slidable from the closed position to an open position through application of a substantially vertical force against the receiving surface and being slidable from the open position to the closed position under the action of biasing means, the closing mechanism in the open position defining a passage between the opening of the bin and a bottom portion of the enclosure located below the closing mechanism, the closing mechanism in the closed position closing the said passage; and the cassette supported by the holder in the

opening of the bin with the clearance allowing the cassette not to impede the motion of the closing mechanism from the closed position to the open position.

(16) Still in accordance with the first embodiment, the holder comprises a first annular flange extending inwardly from an outer wall of the bin, a tubular wall extending generally downwardly from an inner end of the first annular flange, and a second annular wall extending inwardly from a bottom end of the tubular wall, thus forming a seat for receiving the cassette.

(17) Still in accordance with the first embodiment, the tubular wall tapers in a downward direction, and the annular body has a corresponding frustoconical outer periphery for complementary engagement of the cassette in the holder.

(18) Still in accordance with the first embodiment, the opening of the bin is surrounded by a removable lid, the lid opening/closing access to the opening of the bin, the lid being prevented from being installed properly when the cassette is oriented upside down in the holder.

(19) In accordance with a second embodiment of the present application, there is provided: a cassette for dispensing bags from an elongated tubing comprising an annular receptacle accommodating a length of tubing in an accumulated condition, an annular opening at an upper end of the annular receptacle for dispensing the tubing, the annular receptacle defining a central opening through which a knotted end of the tubing passes to form a bag supported by the annular receptacle with disposable objects passing through the circular central opening to be received in the bag, and a clearance only at a bottom of an outer periphery of the annular receptacle.

(20) In accordance with the second embodiment, the clearance forms a frusto-conical geometry at the bottom of the annular receptacle.

(21) In accordance with a third embodiment, there is provided: an apparatus for packing at least one disposable object into an elongated tube of flexible material, comprising: a bin defining an enclosure with an opening defined at a top thereof for receiving at least one disposable object to be packed; a holder within the bin proximate the opening, and an interfering member at a bottom of the holder; a closing mechanism located below the holder comprising first and second portions, the second portion being biased in a closed position against the first portion and defining a receiving surface for receiving the at least one disposable object, the second portion being slidable from the closed position to an open position through application of a substantially vertical force against the receiving surface and being slidable from the open position to the closed position under the action of biasing means, the closing mechanism in the open position defining a passage between the opening of the bin and a bottom portion of the enclosure located below the closing mechanism, the closing mechanism in the closed position closing the said passage, a cassette supported by the holder in the opening of the bin and having an annular receptacle to accommodate a length of tubing in an accumulated condition, the annular receptacle having an upper annular opening for dispensing the tubing such that the tubing extends through a central opening of the annular receptacle and with a knotted end thereof into the enclosure of the bin to receive disposable objects, whereby the closing mechanism closes the tubing, the annular body having a clearance in a bottom portion of an outer periphery of the annular receptacle, so as to cooperate with the interfering member for the cassette to be aligned with the top of the bin in a desired orientation; and a removable lid positioned flush on the top of the bin only when the cassette is in the desired orientation, the cover opening/closing access to the opening of the bin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view of an apparatus according to a first embodiment of the present invention;

(2) FIG. 2A is a bottom perspective view of a cassette to be used with the apparatus of FIG. 1;

(3) FIG. 2B is a top perspective view of the cassette of FIG. 2A;

(4) FIG. 3A is a cross-sectional view of the apparatus of FIG. 1, without tubing, with a closing mechanism in a closed position;

(5) FIG. 3B is a cross-sectional view of the apparatus of FIG. 3A, with the closing mechanism displaced toward an open position;

- (6) FIG. 3C is a cross-sectional view of the apparatus of FIG. 3A, with the closing mechanism in the fully open position;
- (7) FIG. 4 is a cross-sectional view of an apparatus according to a second embodiment of the present invention;
- (8) FIG. 5A is a bottom perspective view of a cassette to be used with the apparatus of FIG. 4;
- (9) FIG. 5B is a top perspective view of the cassette of FIG. 5A;
- (10) FIG. 6 is an enlarged perspective view of an apparatus in accordance with a third embodiment of the present invention; and
- (11) FIG. 7 is a sectional view illustrating the cassette of FIG. 2A as used with an apparatus having interfering members.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (12) Referring to FIG. 1, there is shown an apparatus **10** for packaging disposable objects in an elongated tube of flexible plastic film material. The apparatus **10** comprises a bin **12** having a top portion **14** and a bottom portion (not shown) hingedly connected to each other by a hinge member **16**, with outer walls of the top portion **14** and bottom portion defining an enclosure **18** of the bin **12**.
- (13) The top portion **14** and bottom portion are locked together with a mechanism such as a push button latch (not shown) received in eyelet **20**, located opposite of the hinge member **16**. Upon actuation of the push button, the top portion **14** and bottom portion are disconnected and can be pivoted apart about the hinge member **16** to provide access to the enclosure **18**, for example to empty the bin **12**.
- (14) The top portion **14** defines an opening **22** for receiving disposable objects therethrough. The bin **12** further comprises a lid **24** hingedly connected to a ring **25**, for instance in the shape of a funnel (hereinafter funnel **25**), although multiple shapes are considered such as a flat ring, as long as the ring forms an opening guidance. The funnel **25** is mounted to the top portion **14** such as to selectively open and close access to the opening **22** through pivoting motion of the lid **24**. Although in the embodiment shown the apparatus has a generally circular opening **22**, the bin **12** is not limited to circular openings and could function with openings of different shapes.
- (15) Still referring to FIG. 1, the apparatus **10** also comprises a holder **26** located within the top portion **14** proximate the opening **22** for holding a cassette **30** of flexible plastic film tubing **32**. The cassette **30** has an annular body defining a central opening **34** which is generally aligned with the bin opening **22** when the cassette **30** is accommodated in the holder **26**.
- (16) Referring concurrently to FIGS. 1, 2A and 2B, the cassette **30** has a lid **36** and a bottom annular receptacle **38**. The bottom annular receptacle **38** has a lower surface **31**, and an edge **35** delimiting the lower surface **31**. After the lid **36** is removed, the flexible tubing **32** exits from an upper annular opening of the receptacle **38**, adjacent an outer periphery thereof. It is pointed out that once the lid **36** is removed, there remains an upper flange **39** adjacent to the annular opening that maintains the compressed tubing **32** in the receptacle **38**. The cassette **30** comprises a length of flexible plastic film material that is accumulated (e.g., folded). As shown in FIG. 1, a knot **40** is found at the free end of the tubing **32** to form a bag end. The knot **40** and a length of the tubing **32** extend from the cassette **30** into the enclosure **18** of the bin **12**, passing through the central opening **34** of the cassette **30**. To better contain odors, it is preferable that the film material of the tubing **32** be a multiple layer film having an odor barrier therein such that odors do not permeate through the film material. The receptacle **38** of the cassette **30** defines a chamfer clearance **41** at a bottom of the central opening **34**. The chamfer clearance **41** is provided in order to ensure that the cassette **30** is properly installed in the holder **26** when the apparatus **10** is in use, as will be described hereinafter. The chamfer clearance **41** is provided on the full periphery of the cassette **30**, but may also be partial (i.e., not on the full periphery of the central opening of the cassette).
- (17) Referring to FIG. 1, the holder **26** includes a top annular flange **42** extending inwardly from the top of the bin **12**. A tubular wall **44** extends downwardly from an inner end of the top annular flange **42**. The holder **26** has a horizontal flange **45** at a bottom of the wall **44**, so as to define a seat supporting the cassette **30**. One will also appreciate that the holder **26** may take many alternate shapes and forms.
- (18) Referring to FIGS. 1 and 3A-3C, the apparatus **10** further includes a closing mechanism **50** whose purpose is to provide access to the flexible tubing **32** extending therebelow and preventing odors from escaping from the flexible tubing **32**. In one form of use, it permits the disposable objects to push down against the bottom of the tubing **32** (at the knot **40**) to draw out loose flexible tubing from the receptacle **38** of the cassette **30**. One aspect of note is that the flexible tubing **32** passes over receptacle **38** and the closing

mechanism **50**, thereby installing the receptacle **38** and the mechanism **50** from the disposable objects to be packaged, reducing the need for constant cleaning of the walls defining the opening **34** and the mechanism **50**. In this respect, one will appreciate that one of the preferred uses for the apparatus **10** is the disposal of baby diapers.

(19) The closing mechanism **50** comprises a fixed portion **52**, which comprises a generally “C”-shaped member rigidly connected to the wall **44** of the holder **26**. Alternate types of connections are also possible for the fixed portion **52**, including connection to an additional support member connected to the holder **26** or to another part of the top portion **14**. The curved fixed portion **52** defines a rounded apex **54** which projects into the opening **22**, as can be seen in FIG. **1**. The fixed portion **52** defines a receiving surface **56** downwardly inclined toward a movable portion **58** of the closing mechanism **50**.

(20) The movable portion **58** of the closing mechanism **50** is also generally “C”-shaped in cross-section and is biased against the fixed portion **52** in a closed position shown in FIGS. **1** and **3A** such as to generally close the opening **22** of the apparatus **10**, thereby sealing the tubing **32** shut. The curved movable portion **58** defines a rounded apex **60** projecting into the opening **22**, as can be seen in FIG. **1**. The movable portion **58** defines a receiving surface **62** downwardly inclined toward the fixed portion **52**, the two receiving surfaces **56,62** forming a trough directing an object therebetween. In the closed position, contact between the two portions **52,58** is between the apex **60** of the movable portion **58** and a lower part of the fixed portion **52** located below the apex **54** thereof. These portions are wave-shaped in cross-section to provide an ergonomical shape in view of hands contacting them, as illustrated in FIG. **1**. Moreover, these wave-shaped portions provide efficient sealing of a plastic film tube **32** engaged or pinched therebetween.

(21) The movable portion **58** is slidably supported by a pair of spaced-apart, generally vertical walls **64** (only one of which is shown in FIG. **1**) extending downwardly from the holder **26**, with the opening **22** being aligned therebetween. Each of the walls **64** has an angled slot **66**. In the embodiment shown, the slots **66** are angled at approximately 40 degrees with respect to the horizontal, although any other working angles are suitable.

(22) The movable portion **58** includes aligned sliding members **66A** extending therefrom and engaging the slots **66**, such that the movable portion **58** is slidably movable along an angular direction defined by the slots **66**, and thus along direction A. The movable portion **58** can move between a closed position, as illustrated in FIGS. **1** and **3A**, in which the movable portion **58** abuts against the fixed portion **52** to seal the tubing **32** shut, and an opened position (shown in FIG. **3C**) enough to let a disposable object to pass between the portions **52** and **58**. The movable portion **58** is biased to the closed position of FIG. **1**, for using an elongated helical spring **67** connected to and extending between the sliding members **66A** of the movable portion **58** and around the back of the fixed portion **52**. The concave shape of the back of the fixed portion **52** helps maintain the spring aligned in a plane defined by the slots **66**.

(23) It is of particular importance that the cassette **30** be installed in the appropriate orientation. In the appropriate orientation, as illustrated in FIG. **1**, the tubing **32** is deployed from the top of the cassette **30**. Accordingly, the tubing **32** covers the side walls of the cassette **30** in the opening **34**. Accordingly, as the tubing **32** covers the cassette **30**, there is no risk of contamination of the walls of the cassette.

(24) As is clearly seen in FIG. **1**, the chamfer clearance **41** is positioned downwardly. A path of movement of the movable portion **58** is illustrated at B. It is observed that the movable portion **58** passes closely to the wall defining the chamfer clearance **41**. If the chamfer clearance **41** were not provided, the cassette **30** would impede the movement of the movable portion **58**. Accordingly, if the cassette **30** were installed upside down, the movable portion **58** would be prevented from moving along its path. Therefore, if a user person wants to use the cassette **30** properly, the cassette **30** must be oriented properly.

(25) Alternately, the fixed portion **52** can be replaced by a portion movable opposite of the movable portion **58**, the two being biased together in the closed position. Such a configuration could be, for example, two sliding members symmetrical with respect to one another about a vertical axis defined at the point of contact therebetween, such as two “C”-shaped members similar in configuration to the movable member **58** shown, two rollers, etc., which are slidable away from each other upon downwardly pushing an object thereon, and are biased together for example by being interconnected by one or a plurality of springs. However, such a mechanism includes additional parts.

(26) In order to assist the user in correctly placing the disposable object in the tubing **32**, the apparatus **10** advantageously has the funnel **25** removably installed over the cassette **30**, providing a downward shape around the openings **22,34**, which creates a funnel effect and guides the user's hand. A handle **68** is

provided to remove the lid **24**/funnel **25** assembly.

(27) Referring to FIG. **1**, the apparatus **10** can further include, within the bin **12**, an integrated cutting tool **70**, which can be for example attached to one of the vertical walls **64**, and which includes a blade located between two spaced-apart protective fingers sized such as to prevent a user's finger from reaching the blade but still allowing a portion of tubing **32** to be inserted between the fingers to be cut with the blade.

(28) In order to prepare the apparatus **10** for use, and with reference to FIG. **1**, a user person removes the lid **24** and funnel **25** assembly and inserts the cassette **30** in the correct orientation (i.e., with the chamfer clearance **41** facing downward. The user person withdraws a free end of the tubing **32** from the cassette **30**, takes a leading edge of flexible tubing **32**, ties it into the knot **40** and pushes the knot **40** through the openings **22,34** and beyond the closing mechanism. It is pointed out that the knot **40** may be made prior to the cassette **30** being installed. The movable portion **58** opens as a result of the downward pressure, whereby the knot **40** reaches the enclosure **18** below the closing mechanism **50**.

(29) Alternately, the tube end with the knot **40** could be pulled down to the bottom of the bin **12**, for example by pivoting and opening the top portion **14** such as to gain access to the enclosure **18**, and pulling the tube end with the knot **40** through the closing mechanism **50** and beyond to the bottom of the bin **12**. The top portion **14** is then re-engaged with the bottom portion, and the lid **24**/funnel **25** assembly is reinstalled on the top portion **14**. Accordingly a length of empty flexible tubing **32** extends to the bottom and is now ready to receive disposable objects therein. Therefore, the film material is not pulled from the cassettes **30** as disposable objects are placed into the tube.

(30) In use, when a disposable object needs to be packaged, the lid **24** of the apparatus **10** is lifted, thereby providing access to the opening **22**. The disposable object to be packaged is placed into the opening **22** (in this case, the cassette **30** of flexible tubing **32** surrounds the object). The receiving surfaces **56,62** of the fixed and movable portions **52,58** of the closing mechanism **50** biased together in the closed position form a trough which directs the disposable object in the hand of a user person toward a contact region therebetween.

(31) The user person pushes the disposable object in a downward direction, gradually opening the movable portion **58**, as is shown in FIGS. **3B** and **3C**. The angled receiving surface **62** of the movable portion **58** helps the downward force to be easily converted along the direction of the slots **66** such as to force the movable portion **58** from its closed position in contact with the fixed portion **52** (FIG. **3A**) to its open position (FIG. **3C**) against the force of the spring **67**, through a sliding movement illustrated by direction A in FIG. **1**. Since the movable portion **58** is moved by the force applied on the disposable object, it slides only enough for the disposable object to go past the closing mechanism **50**, with most of the opening created by the movement of the movable portion **58** being blocked by the disposable object such as to minimize the propagation of odors through the opening **22**.

(32) Once the disposable object passes beyond the apex **60** of the movable portion **58**, the user person releases the disposable object and withdraws his or her hand. The movable portion **58**, under the action of the spring **67**, rapidly returns to the closed position of FIGS. **1** and **3A**, through a sliding movement, effectively sealing the object within the flexible tubing **32**. The curved profile of the fixed portion **52** below the apex **60** and, more importantly, of the movable portion **58** below the apex **60** helps push the disposable object further down, if there is resistance, as the movable portion **58** returns to the closed position to apply a downward pushing force.

(33) It is pointed out that different configurations of the cassette are considered. For instance, clearances of different shapes can be used instead of a chamfer. The use of a chamfer shape does not substantially reduce the volume of the receptacle **38** of the cassette **30**, thereby generally preserving the quantity of tubing **32** received in the cassette **30**.

(34) In an alternative embodiment illustrated in FIGS. **4**, **5A** and **5B**, the cassette **30'** is provided with a tapered clearance **41'** (forming a frustoconical geometry at the outer periphery of the bottom of the cassette **30'**) as opposed to a chamfer clearance **41**. The tapered clearance **41'** is used in conjunction with a tapered flange **44'** in the holder **26** of the apparatus **10**, as is clearly illustrated in FIG. **4** (i.e., the flange **44'** defines an interference member of frustoconical shape). Accordingly, this complimentary shape ensures that the cassette **30'** is properly oriented in the holder **26**, otherwise the funnel **25** could not be installed properly on the top of the bin **12**. Moreover, the lid **24** and funnel **25** assembly would have problems closing with the improper orientation of the cassette **30'**. It is pointed out that like reference numerals refer to like elements in FIGS. **1** and **4**.

(35) For each new disposable object to be packed, the process is repeated, each time either dropping the disposable object within the already drawn length of flexible tubing **32** within the enclosure **18**, or drawing out an additional length of flexible tubing **32** from the cassette **30** to accommodate another disposable object as the disposable object moves down into the tube inside the apparatus **10**, until the enclosure **18** is full, containing an object-filled segment of flexible film tubing **32**. In this case, the top portion **14** is pivoted away from the bottom portion. The flexible tubing **32** is then cut, for example by using the cutting tool **70** provided, and then tied to form a substantially sealed pouch which can be removed from the bin **12**. The top portion **14** is pivoted back into engagement with the bottom portion to use the apparatus **10** again as described above.

(36) In another embodiment of the apparatus **10** using the cassette **30** of FIGS. 2A and 2B, the fixed portion **52'** of the closing mechanism **50** has an upper end **54'** that extends upwardly in the opening **22** of the bin **12**. The upper end **54'** is sized such that it also extends into the opening **34** of the cassette **30** when the cassette **30** is positioned in the holder **26**. If it were positioned upside down, the cassette **30** would extend above the annular flange **42**, and the ring **68** would not be positionable on the bin **12**.

(37) In another embodiment illustrated by FIG. 7, the holder **26** features an interfering member **70** that has a shape that is complementary to that of the cassette **30** with the chamfer clearance **41**. As shown, the interfering member **70** has an oblique cassette-orienting wall **71** supporting the cassette **30**. The edge **35** of the bottom annular receptacle **38** of the cassette **30** engages the oblique cassette-orienting wall **71**. With the presence of the interfering member **70**, the cassette **30** must properly be installed in the holder **26** to be used. As illustrated in FIG. 7, a pair of interfering members **70** may optionally be provided on both side of the holder **26**.

(38) Advantages of the apparatus **10** include the use of a wedge-type closing mechanism **50** of simple construction with few parts to seal the flexible tubing **32**. The cassette configurations ensure that the cassette **30** is properly installed in the holder **26**, thereby preventing contamination of the walls of the cassette (and the hands of the user person) as the properly installed cassette is mostly covered by the tubing **32** as is shown in FIG. 1. The mechanism **50** greatly reduces the complexity of the apparatus **10**, makes the assembly simple, and allows for the use of a lesser quantity of flexible tubing **32** than prior-art devices. Furthermore, angling the course of the movable portion **58** permits the optimization of little horizontal clearance versus the force required to bias the movable portion **58**. Finally, the shape of the individual portions of the mechanism prevents disposable objects from “coming back up”—that is, once a disposable object has been inserted under pressure to pull film material, it will be forced to remain in the flexible tubing by the closing mechanism **50**.

Claims

1. A method for disposing of objects in a pail of a waste disposal system having a film-dispensing cassette, the film-dispensing cassette including an annular receptacle defining a central opening, the annular receptacle including a volume receiving an elongated tube of flexible material radially outward of the central opening, the method comprising: orienting the annular receptacle so as to have a bottom of the annular receptacle facing a cassette holder located in an upper portion of the pail; seating a portion of the annular receptacle of the film-dispensing cassette in the cassette holder against an interfering member of the cassette holder having an oblique cassette-orienting wall supporting the annular receptacle, with part of the elongated tube of flexible material passing through the central opening of the annular receptacle and extending into an enclosure of the pail; and disposing waste into the elongated tube of flexible material that extends into the enclosure of the pail through the central opening of the annular receptacle.
2. The method of claim 1, comprising, folding the elongated tube of flexible material over an upper flange attached to the annular receptacle to pass the part of the elongated tube of flexible material through the central opening of the annular receptacle.
3. The method of claim 1, further comprising opening a lid of the pail to gain access to the cassette holder.
4. The method of claim 1, further comprising aligning the central opening of the annular receptacle with an opening of the cassette holder.
5. The method of claim 1, further comprising forming a bag end by knotting a free end of the elongated tube of flexible material.
6. The method of claim 1, comprising further withdrawing the elongated tube of flexible material from the

volume of the annular receptacle.

7. The method of claim 1, further comprising cutting the elongated tube of flexible material, and forming a substantially sealed pouch by tying a cut end of the elongated tube of flexible material.
 8. The method of claim 7, further comprising removing the substantially sealed pouch from the pail.
 9. A method for installing a film-dispensing cassette in a pail of a waste disposal system, the film-dispensing cassette including an annular receptacle defining a central opening, the annular receptacle including a volume receiving an elongated tube of flexible material radially outward of the central opening, the method comprising: withdrawing a length of the elongated tube of flexible material from the volume of the annular receptacle, the length of the elongated tube of flexible material including a free end of the elongated tube of flexible material; passing the free end through the central opening of the annular receptacle; and seating the annular receptacle against an interfering member of a cassette holder in an upper portion of the pail, the interfering member having an oblique cassette-orienting wall supporting the annular receptacle and engaging an edge of the annular receptacle, the edge delimiting a lower surface of the annular receptacle.
 10. The method of claim 9, wherein the withdrawing of the length of the elongated tube of flexible material from the volume of the annular receptacle includes pulling on the free end of the elongated tube of flexible material out from an upper annular opening of the annular receptacle, the upper annular opening defined atop the volume of the annular receptacle.
 11. The method of claim 9, wherein the withdrawing of the length of the elongated tube of flexible material from the volume of the annular receptacle includes pulling the free end of the elongated tube of flexible material into an enclosure of the pail.
 12. The method of claim 9, further comprising forming a bag end by knotting the free end of the elongated tube of flexible material.
 13. The method of claim 12, wherein the forming of the bag end is achieved prior to the passing of the free end of the elongated tube of flexible material in the central opening of the annular receptacle.
 14. The method of claim 9, further comprising aligning the central opening of the annular receptacle with an opening in the upper portion of the pail.
 15. The method of claim 9, further comprising opening a lid of the pail to gain access to the cassette holder.
 16. The method of claim 9, comprising, folding the length of the elongated tube of flexible material over an upper flange attached to the annular receptacle by the passing of the free end of the elongated tube of flexible material through the central opening of the annular receptacle.
 17. The method of claim 9, wherein the seating of the annular receptacle includes abutting an oblique wall of the annular receptacle against the interfering member.
 18. The method of claim 9, wherein the withdrawing of the length of the elongated tube of flexible material from the volume of the annular receptacle is achieved prior to the passing of the free end of the elongated tube of flexible material through the central opening, the passing of the free end being achieved prior to the seating of the annular receptacle against the interfering member of the cassette holder of the pail.
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